

Data quality, distributions, descriptive evidence

Quantitative Methods for Business Analysis. The essentials of every section in one short read: ideas, worked examples and checks.

≈ 27 min read

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Full edition

1 · READ

Open a section, read its central idea, then its worked example.

2 · APPLY

Try the guided practice and the lab task linked in each section.

3 · CHECK

Answer the study check in the writing space without looking back.

Your study route

3.1

Data quality, provenance and measurement scales

7 blocks · slides 10–26 · Part A

3.2

Describing distributions: center, spread and shape

7 blocks · slides 27–45 · Part B

3.3

Visual evidence and honest displays

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Grouped summaries and lab synthesis

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3.1 · Data quality, provenance and measurement scales

Learning purpose: Audit the meaning, origin and eligibility of records before calculating.

CENTRAL IDEA

A calculation is only as defensible as the meaning, origin and structure of the records entering it.

IN CLASS

[Slides 10–26](#)

The Meaning of Measurement Scales in Business... 12

Mathematical Admissibility by Measurement Scale 13

SPSS and Excel Scale Validation Syntax 14

What Can We Calculate? 15

What Does One Row Represent? 16

PRACTISE IT IN THE LAB

- [Part A: Can we trust the order file?](#) · 6 min · SPSS H1 Scale and Sentinel Audit

About 7 min read · PDF pages 4–8 · [Full explanation \(Full Toolkit\)](#)

Begin with the record

A dataset is a collection of recorded values. An **observation** is one recorded case, such as an order; a **variable** is an attribute that can differ across cases, such as delivery time. Before calculating, write what one row represents. This is the table's **grain**. One row per order differs from one row per product within an order. Counting rows produces an order count only when that grain supports it.

The management question determines which observations belong in the analysis. A **denominator** is the eligible count used in an average, proportion or rate. For average completed-delivery time, use valid completed durations. An order still travelling has no completed duration yet. Recording it as zero would mean instantaneous delivery, which is a different claim.

Provenance records where data came from and how it changed: source file or system, extraction date, reporting period, owner, definitions and transformations. A transformation is a documented change such as converting hours to minutes. Keep the original source and record the rule separately. This allows another analyst to trace a result back to its inputs.

Let meaning determine the calculation

The **measurement scale** describes which comparisons have a meaningful interpretation. A number stored in a cell does not automatically measure an amount.

The **mean** adds quantitative values and divides by their count. The **median** identifies the middle position after ordering. Section 3.2 explains both. An order identifier uses digits to distinguish records; averaging identifiers does not count orders or describe their value. For ordinal categories, report the category distribution and meaningful positions. Subtracting arbitrary category codes does not create a measured distance or a meaningful numerical spread.

A **data dictionary** defines each field, units, allowed codes and missing-value rules. Consult it before treating an unusual value as an error. A **sentinel** is a special code representing a condition, such as an unknown age, rather than the measured quantity. Sentinel rules belong to particular variables. The same number can be legitimate in another field.

Scale	What the values mean	Appropriate starting summaries
Nominal	Labels without a natural order, such as hub names	Counts, proportions and mode, the most frequent category
Ordinal	Ordered categories whose gaps are not known to be equal, such as service tiers	Frequencies and ranked positions, including the median category
Interval	Equal differences, but zero is not absence of the quantity, such as Celsius temperature	Means and differences; describe variation when meaningful
Ratio	Equal differences and a meaningful zero, such as elapsed minutes	Means, differences and meaningful ratios, subject to the analytical question

Follow a defensible audit sequence

1. **Name the decision and eligible population.** The population is the set of cases the question concerns. State the period and what counts as an included case.

2. **Confirm the source and grain.** Identify the file version and the field or combination of fields that should distinguish observations.
3. **Check repeated identifiers.** A repeated customer identifier may represent several valid orders. A repeated order identifier may represent product lines. Investigate before removing anything.
4. **Check each relevant variable.** Review type, units, categories, plausible range, blanks and documented sentinels. A blank means no value was recorded; its cause still needs investigation.
5. **Separate validation from correction.** Validation tests a rule. Correction changes data. Document evidence for a correction and preserve the original extract.
6. **Recalculate the denominator.** State the source row count, relevant exclusions and valid count for each summary. Missing age need not exclude an otherwise valid duration.
7. **Record the unresolved limitation.** State which assumption could change the conclusion and what evidence would resolve it.

Do not remove a row merely because it looks inconvenient. A duplicate, a missing field and a rare valid event require different responses. Likewise, do not overwrite a suspicious duration with an average simply to make a chart look orderly.

Worked audit: seven transaction records

This small teaching example uses days and has its own age-code rules. It is separate from the issued Metro lab, which records latency in minutes.

Order	Customer age	Delivery days	What needs checking
ORD1001	34	3	No identified issue
ORD1002	-1	4	Documented unknown-age code
ORD1003	28	Blank	Order still in transit
ORD1004	45	2	Original order record
ORD1004	45	2	Verified repeat of the preceding record
ORD1006	Blank	5	Age unrecorded
ORD1007	999	3	Documented age non-response code

The intended grain is one order per row. Source investigation confirms that the second ORD1004 row is an import repeat. Retain one copy for analysis and document the correction. Six unique orders remain. Five have completed durations: **3, 4, 2, 5 and 3 days**. Their sum is 17 days, so mean completed duration is $17 \div 5 = 3.4$ days.

The sixth unique order is still in transit. It is excluded from this completed-duration mean, not declared an invalid order. The finding describes five completed orders in this small extract; it does not establish the experience of all orders, including the unfinished one. Age codes do not change those five valid delivery durations.

GO DEEPER Deeper explanation: why missingness matters

Missingness is the process through which values become unavailable. The proportion missing tells you its size, but not its effect. Three standard terms describe different assumptions about that process.

Missing completely at random (MCAR) means the chance of a value being missing is unrelated to both observed information and the unavailable value. Random scanner failure could fit this assumption if failure is unrelated to the order or its duration. The remaining observations may still represent the target reasonably, but there are fewer of them.

Missing at random (MAR) means that, after accounting for observed information, missingness no longer depends on the missing value. For example, an older intake form used in one recorded region might omit a field. Adjustment could use the observed region, but its adequacy remains an assumption. “At random” does not mean the blanks have no explanation.

Missing not at random (MNAR) means missingness still depends on the unavailable value after observed information is considered. Customers with very high incomes might be less likely to disclose income. The values most relevant to checking the assumption are missing, so the observed table alone usually cannot settle it.

Compare missingness across time, location and collection processes; ask the data owner what happened. **Imputation** replaces missing values with estimates under stated assumptions. It is not an automatic cleaning step in this lab. There is no universal missing-percentage cutoff that makes every analysis trustworthy or unusable. Explain what is missing and whether plausible missing values could change the decision.

Apply the audit to the issued lab

Open `LAB01_Learning_Workbook.xlsx`, begin on **Start Here**, and save your personalized copy. The **Data** sheet contains the Excel table **Orders**, with 10,000 recorded orders. An Excel Table is a named range whose column names can be used in formulas. Keep these source rows.

In **IN-CLASS_Lab Part A**, cells **B7:B9** contain the scheduled sentinel counts and scale choice. The issued rules are `Age_Years = -1` for missing age and `Latency_Minutes = 999` for missing latency. Unlike the small example above, the lab does not declare 999 an age sentinel. Follow each field's own rule.

COUNTIF counts cells meeting one condition; **COUNTIFS** counts rows meeting several conditions together. These functions count recorded matches. They cannot determine the meaning of a code without the dictionary. Read the adjacent workbook check, then explain the result in the designated response area.

IBM SPSS Statistics originated as the **Statistical Package for the Social Sciences (SPSS)**. Here it is the second software route for checking the analysis. During the scheduled evidence clinic, run the supplied **H1 Scale and Sentinel Audit** block and record its output title. The lab guide supplies the Mac screenshots and exact steps; this toolkit adds no submission task.

Check your reasoning

Use this approach when receiving a new extract, joining sources or comparing reporting periods. Pause interpretation when the grain or code meanings are unresolved. A calculation can execute successfully while answering the wrong question.

Reflect: Management wants a result today, but one region's collection rules are unclear. Reporting now may misrepresent that region; waiting delays action. Which would you recommend, and what consequence would you monitor?

Recall without looking back: Why can a record with missing age still belong in a delivery-time mean? Name the denominator you would report.

Lab connection: Complete Part A's existing cells, reason and H1 caption when scheduled. Use the guide's checkpoint and save instructions before moving to Part B.

3.2 · Describing distributions: center, spread and shape

Learning purpose: Choose, calculate and interpret summaries that match the management question.

CENTRAL IDEA

No single statistic describes the typical experience, variation and extreme outcomes at once.

IN CLASS

[Slides 27–45](#)

What Measures of Center Mean 24

Measures of Central Tendency 25

How Resistance and Breakdown Points Operate
26

Spread and Distribution Shape 27

Sample Variance and Sample Standard
Deviation... 28

PRACTISE IT IN THE LAB

- [Part B: What does a typical order look like?](#) ·
7 min · SPSS H2 Center and Spread

About 7 min read · PDF pages 9–13 · [Full explanation \(Full Toolkit\)](#)

Start with the distribution

A **distribution** describes the values a variable takes and how often they occur. Read it through three questions: Where do values gather? How much do they vary? What pattern do they form? These are **center**, **spread** and **shape**. Two processes can have the same center but very different customer experiences.

A **key performance indicator (KPI)** is a measure chosen to track an important objective. Calling a number a KPI does not determine its statistical meaning. A delivery KPI might be a median, mean or proportion meeting a target. Define the indicator's calculation and eligible records before interpreting improvement.

The **population** is the complete set of cases the question concerns. A **sample** is a subset observed to learn about it. A file can include every recorded order for one period yet still be an incomplete basis for predicting future periods. This session emphasizes **descriptive statistics**, which summarize the

observations actually analyzed. A descriptive difference alone does not establish its cause or future stability.

Choose the center for the question

Summary	Meaning	Useful question	Limitation
Mean	Sum of valid values divided by valid count	What is the average numerical burden per observation?	Distant values can move it substantially
Median	Middle ordered value, or average of the two middle quantitative values	What is the middle recorded experience?	Does not show how large the extremes are
Mode	Most frequent value or category	Which category or value occurs most often?	Several values can tie; frequency needs context

The mean uses every valid quantity. That makes it useful for understanding totals, including costly extremes. The median is **resistant**: a small number of distant observations usually changes it less. Resistance is not immunity. If enough observations change, the median changes too. Neither summary is universally best.

Delivery minutes are elapsed time. Their sum describes accumulated duration across orders; it does not automatically equal paid labor hours because orders may move at the same time. Connect the statistic to the business quantity it actually measures before using it for a staffing or cost decision.

Understand spread and shape

The **range** is maximum minus minimum. It is easy to read but depends entirely on two observations. A **percentile** is a position-based cut point in ordered data. Estimated cut points can lie between observed values. Ties and calculation conventions mean an exact stated percentage need not fall strictly below the returned number.

The **first quartile (Q1)** estimates the 25th-percentile cut point; the **third quartile (Q3)** estimates the 75th. The **interquartile range (IQR)** is Q3 minus Q1, describing the width of the middle half. For numerical durations, it remains in minutes. Arbitrary ordinal category codes do not support meaningful subtraction merely because categories can be ordered.

Variance measures spread using squared distances from the mean. **Standard deviation (SD)** is its square root and returns to the original measurement unit. It is a useful distance-based summary, but not the arithmetic average of absolute distances. Large deviations receive strong weight because they are squared.

Skewness describes asymmetry. A longer high-value tail indicates right skew; a longer low-value tail indicates left skew. The mean often lies toward the longer tail, but the mean-median relationship is a clue, not a universal diagnostic rule. Equal values do not prove symmetry. **Kurtosis** concerns tail weight and the influence of extreme observations; treating it only as the height of a peak misses its practical meaning.

Worked example: ten fulfillment durations

Use the same small teaching sample as the slide example. One observation is one shipment from a single-shift audit. Durations, already ordered, are 22, 24, 25, 26, 28, 29, 31, 33, 48 and 74 minutes. These are separate from the 10,000-row lab extract.

Let n be the valid sample count, x_i one duration and $\bar{x}$ the sample mean. The symbol Σ means add the indicated values. Here $n = 10$ and $\Sigma x_i = 340$ minutes. Therefore:

Mean: $\bar{x} = \Sigma x_i / n = 340 / 10 = 34.0$ minutes
Median: $(28 + 29) / 2 = 28.5$ minutes

To calculate sample spread, subtract 34 from each duration, square each distance and add the results. The first five squared distances total 425; the remaining five total 1,831. Their combined sum is **2,256 square minutes**. Let s^2 denote sample variance and s sample SD.

$s^2 = \Sigma (x_i - \bar{x})^2 / (n - 1) = 2,256 / 9 = 250.67 \text{ min}^2$
 $s = \sqrt{250.67} = 15.83$ minutes

The square-root sign, $\sqrt{}$, reverses squaring for a non-negative value. Keep the squared unit for variance; report minutes for SD. The 74-minute observation contributes $(74 - 34)^2 = 1,600 \text{ min}^2$, most of the total squared distance. Its influence is real even though only one shipment has that duration.

Using the lab's **exclusive quartile** convention, $Q1 = 24.75$ and $Q3 = 36.75$ minutes, so **IQR = 12.00 minutes**. “Exclusive” names Excel's `QUARTILE.EXC` calculation method. With ten observations, the first-quartile position is $(10 + 1) \times 0.25 = 2.75$. Move three quarters of the way from the second value, 24, to the third, 25: 24.75. The same interpolation gives $Q3$ between 33 and 48. **Interpolation** estimates between adjacent values.

The sample therefore has a mean of 34, median of 28.5, SD of 15.83 and IQR of 12 minutes. The unusually long 74-minute duration matters to the upper tail. These figures describe the sample; they do not identify why that shipment took longer.



Once the sample mean is calculated, deviations from it must sum to zero. If nine deviations and the mean are fixed, the tenth deviation is determined. This leaves **nine degrees of freedom**, meaning nine independently variable deviations. Dividing by n minus one corrects the downward bias of the variance estimate under standard random-sample assumptions. It does not correct poor sampling or missing records.

If describing an entire defined population, the population-variance formula instead divides by its population count. Choose sample or population deliberately; a software default is not a research design. The supplied lab requests sample SD, so use `STDEV.S` in Excel and the corresponding sample statistic in SPSS.

Observed variation is also different from uncertainty about an estimated mean. SD describes differences among individual observations. A **confidence interval**, studied later, expresses uncertainty about a population quantity estimated from data. A precise mean does not imply that individual deliveries are similar or that every future delivery will finish near it.

A short analysis workflow

1. Confirm the unit, period, grain and valid count. Apply missing-value rules before calculation.
2. Inspect ordered values or a distribution chart for tails, clusters and suspicious records.
3. Calculate a meaningful center and spread. Compare mean/SD with median/IQR when extremes matter.
4. Check the percentile convention. Different methods may produce different quartiles on the same small sample.
5. Write the result with units and a limitation. Describe what changed before claiming why.

Use mean and SD when their arithmetic answers the question; skewness does not make them mathematically invalid. Use median and IQR to describe the central experience with less influence from distant values. Report both when their contrast is useful. A service target, or **service-level agreement (SLA)** commitment, requires business context as well as statistical summaries.

Connect the calculation to Part B

In IN-CLASS_Lab Part B, complete B7:B10 for mean, median, sample SD and IQR. The supplied `FILTER` conditions retain valid latency while excluding 999 and blanks. `AVERAGE`, `MEDIAN`, `STDEV.S` and `QUARTILE.EXC` summarize that filtered set. Do not substitute a different quartile method simply because its

output looks close.

The workbook's adjacent check helps verify the calculation. If a number differs, inspect the variable, filter, function and range before retyping an expected value. During the evidence clinic, use **H2 Center and Spread** in SPSS. Compare valid N, meaning the valid observation count, before comparing numerical summaries. Record the designated caption and explanation in the existing fields.

Reflect: A manager wants to publish only the median because it looks reassuring. Including the mean may worry customers, while omitting it hides costly extremes. What would you publish, and why?

Recall without looking back: Why do variance and SD have different units? Why can mean and median both be correct yet answer different questions?

Lab connection: Save Part B's calculation checkpoint and use its IQR in Part C. Keep units, missing-value rules and quartile method consistent across both parts.

3.3 · Visual evidence and honest displays

Learning purpose: Choose and interpret a chart that preserves the comparison and its limits.

CENTRAL IDEA

A useful display makes the evidence easier to inspect without making the claim stronger than the data supports.

IN CLASS

[Slides 47–67](#)

Visual Choice and Distribution Geometry 38

Tukey Outlier Fence Boundaries 39

SPSS and Excel Formulas for Distribution Analysis 40–41

Chart Geometry and Baselines 42

Cognitive Load and Visual Clutter 43

PRACTISE IT IN THE LAB

- [Part C: Which orders are unusually slow?](#) · 6 min · SPSS H3 Tukey and Boxplot Evidence

About 7 min read · PDF pages 14–18 · [Full explanation \(Full Toolkit\)](#)

Choose the picture from the question

A chart represents data using **visual encodings**: properties such as position, length, color or area. An **axis** is a reference scale; its labels tell you what position means. A **legend** explains colors or symbols. An **annotation** is a short explanatory note attached to a relevant feature. These elements should help the reader answer a question, not merely make the chart look finished.

Begin by naming the question and the type of variable. A categorical variable groups observations by labels; a quantitative variable measures an amount. Then choose a display that preserves the comparison you need.

Question	Useful display	What to inspect
How frequent are categories?	Bar chart	Counts or proportions, category labels and denominator
How is one numerical variable distributed?	Histogram	Concentration, spread, tails, gaps and possible clusters
How do numerical distributions differ across groups?	Side-by-side boxplots	Medians, middle-half widths, whiskers and separate points
How does a measure change through ordered time?	Line chart	Timing, direction, changes and gaps in observation

A **histogram** groups numbers into intervals called **bins**. Its horizontal scale is quantitative. A bar chart compares separate categories; their order is not automatically a numerical continuum. A **line chart** connects observations in a meaningful order. Connecting unrelated hub names implies a path that does not exist.

Bar length encodes magnitude, so a bar chart normally begins at zero. A **baseline** is the reference from which that length is judged. Shortening the axis can exaggerate a small difference. Other displays, such as line charts, can use a restricted range when clearly labeled and appropriate to the question. The rule follows the encoding, not a demand that every chart have identical axes.

Read a boxplot step by step

A **boxplot**, also called a box-and-whisker plot, summarizes ordered numerical values. The line inside the box marks the median. The box edges represent the **first quartile (Q1)** and **third quartile (Q3)**, near the 25th and 75th percentile positions. Their difference is the **interquartile range (IQR)**, the numerical width of the middle half.

Tukey fences are calculation boundaries used to flag distant values. Under the common $1.5 \times \text{IQR}$ rule, the lower fence is $Q1 - 1.5 \times \text{IQR}$ and the upper fence is $Q3 + 1.5 \times \text{IQR}$. A **whisker** extends to the most extreme observed value still within its fence. The whisker endpoint is therefore an observation, not necessarily the fence itself. Values beyond the fences appear separately.

An **outlier** is an unusually distant observation under a stated rule. That is a flag for investigation, not proof of error. A genuine delay can be precisely the service failure that management needs to understand. Deleting it solely to improve a chart removes relevant evidence.

Worked example: Metro's Hub C

The issued teaching extract contains 10,000 recorded orders and 9,958 valid latency observations across three hubs. Latency means the recorded elapsed delivery time, measured in minutes per order. Hub C has 3,320 valid latency observations. Use the same valid-record rule and exclusive quartile method as the workbook.

Hub	Q1 (minutes)	Median (minutes)	Q3 (minutes)	IQR (minutes)
A	19	22	26	7
B	19	22	26	7
C	14	22	31	17

All three medians equal 22 minutes, but Hub C's middle half covers a wider interval. A chart showing only median bars would hide this difference in spread. Side-by-side boxplots preserve both the shared center and different variation.

For Hub C, calculate the boundaries in order:

```
IQR = Q3 - Q1 = 31 - 14 = 17 minutes
Fence distance = 1.5 × 17 = 25.5 minutes
Lower fence = 14 - 25.5 = -11.5 minutes
Upper fence = 31 + 25.5 = 56.5 minutes
```

The observed maximum is **124 minutes**, beyond the upper fence. Keep the calculated lower fence of **-11.5 minutes**. Zero is the separate physical lower bound for a duration; it does not replace the statistical calculation. A negative observed duration would require a validity check even though the computed lower fence is negative.

The supported conclusion is that Hub C has wider middle-half variation and some unusually long recorded durations. The chart does not establish that its carrier caused the difference. Route mix, demand, handoffs and recording conditions are possible explanations to investigate. **Attribution** means assigning an outcome to a cause; a descriptive chart alone cannot settle attribution.

GO DEEPER Deeper explanation: every summary hides something

A boxplot is compact because it compresses many observations. That helps compare groups but can hide multiple clusters. A **mode** is a frequent value or concentration; a distribution with several clusters may be **multimodal**. Two quite different distributions can have similar medians and quartiles.

Inspect a histogram or individual observations when the hidden shape matters.

A histogram also makes choices. Narrow bins reveal detail but can look irregular; wide bins smooth the display and may conceal gaps. Try more than one reasonable bin width before treating a feature as stable. Keep the same bins and scales across groups when comparing their shapes. Changing the construction between groups can create an apparent difference.

Software conventions matter too. Some boxplots use **hinges**, position-based box edges calculated by a convention that can differ from an interpolated percentile. The lab uses Excel's `QUARTILE.EXC` for its reported quartiles. A software plot may use another box-edge rule. Check the method and denominator before calling a small discrepancy a mistake. Do not change the issued calculation to imitate a picture.

Direct attention without distorting evidence

Cognitive load is the mental effort needed to understand the display. A useful chart reduces unnecessary effort while retaining the context needed to judge its claim. **Preattentive attributes** are visual features noticed quickly, such as a strong contrast or distinct position. Use one intentional highlight to direct attention to the comparison that matters.

Gestalt principles describe how viewers group visual elements. Proximity makes nearby items seem related. Similarity makes matching colors or shapes seem to belong together. Enclosure places a boundary around a group. Use these relationships deliberately; a box or color can imply a relationship even when the data does not support it.

The **data-ink ratio** expresses the share of a display's ink or pixels devoted to data rather than decoration. It is a design guide, not a score to maximize blindly. A light gridline, target line or explanation may be essential even if it is not itself a data mark. Remove an element only when the reader loses no useful meaning.

A six-step visual check

1. **Read the frame:** question, population, period, units, valid count, source and filters.
2. **Decode the marks:** identify what position, length, color and size represent.
3. **Describe first:** state center, variation and unusual values before proposing causes.
4. **Check comparability:** align units, axes, bins, denominators and quartile conventions.
5. **Focus attention:** use readable labels, one useful highlight and restrained gridlines.
6. **Qualify the claim:** name a limitation or additional check that could change the interpretation.

Use direct labels where they reduce repeated movement to a legend. Do not rely on red and green alone; combine color with position, shape or text. **Alternative text** gives readers using assistive technology a short account of the chart's purpose, main pattern and important exception. The accompanying table can supply exact values.

Avoid three-dimensional decoration that distorts length or area. State whether a bar represents a count or percentage: “how many” and “what share” answer different questions. A title can identify the observed pattern, but should not turn an association into a causal claim. “Hub C has wider observed spread” is supported here; “Hub C's carrier caused delays” is not.

Connect the display to Part C

In **IN-CLASS_Lab Part C**, complete **B7:B11** for fences, the upper-tail count and the two largest valid observations. Read the guide beside the workbook. Its adjacent checks confirm numerical relationships; your explanation should state what a flag means and what it cannot prove.

During the evidence clinic, run **H3 Tukey and Boxplot Evidence** in the supplied SPSS syntax. Inspect the Extreme Values table and case labels. The workbook uses an editable XY chart, placing points on horizontal and vertical coordinates, to draw the boxplot components; it is not Excel's native Box & Whisker chart. Record the existing caption and interpretation when scheduled.

Reflect: Highlighting Hub C could prompt a useful investigation but unfairly imply blame. Leaving every hub visually equal might hide a service concern. What title and emphasis would preserve both urgency and fairness?

Recall without looking back: How does a whisker endpoint differ from a fence? Why should a 124-minute observation remain visible while its cause is investigated?

Lab connection: Save Part C, retain the valid-record rules, and use Part D to compare the three hubs on a shared scale.

3.4 · Grouped summaries and lab synthesis

Learning purpose: Build comparable group summaries and explain what their differences can establish.

CENTRAL IDEA

Reliable comparisons combine clear row meaning, consistent denominators and attention to variation within groups.

IN CLASS

[Slides 68–86](#)

Stacked Data: One Observation per Row 55

Subgroup Sample Mean and Pooled Variance 57

SPSS Means and Group Breakdown Syntax 58

How a PivotTable Groups Records 59

A Descriptive Audit 60

PRACTISE IT IN THE LAB

- [Part D: Is Hub C really different?](#) · 7 min · SPSS H4 Hub Comparative Audit
- [Take-home: Should we penalize Hub C, and what should the SLA say?](#) · 60 min

About 7 min read · PDF pages 19–23 · [Full explanation \(Full Toolkit\)](#)

Keep group membership beside the observation

A **group** is a subset sharing a characteristic, such as a hub or service category. A **dimension** is a field used to organize observations into groups. A **measure** is a numerical variable being summarized. **Aggregation** combines observations into a count, sum, mean or other summary. These terms describe what a grouped table does, regardless of software.

Long, or **stacked**, data places repeated observations on rows and stores group membership in a field. **Wide**, or **unstacked**, data places groups or repeated occasions in separate columns. The best structure depends on the task. For the grouped summaries here, stacked data makes it easier to select a category, summarize its valid observations and repeat the same operation for other categories.

Consider a clinic table with `Patient_ID`, `Triage_Level` and `Wait_Minutes`. **Triage** groups patients by urgency. In this teaching example, Urgent, Priority and Routine are illustrative labels, not a clinical standard. One row represents one patient encounter. The identifier distinguishes encounters; the category identifies the group; the duration supplies the measurement.

See how the grouping works

Data structure	Example organization	Practical consequence
Wide	Separate Urgent_Wait, Priority_Wait and Routine_Wait columns	Group identity is stored in the column name; unequal counts leave uneven columns
Stacked	One Wait_Minutes column plus a Triage_Level column	The same calculation can be repeated for each category

A **PivotTable** is an Excel summary that groups records by selected fields and applies a calculation within each group. Putting Triage_Level in **Rows** creates one output row per category. Putting Wait_Minutes in **Values** tells Excel which measurement to summarize. The selected summary function determines whether you see a count, sum, average or something else.

Check the result's label. Excel may initially choose Sum or Count based on the stored values. A number-looking field can contain text, and blank measurements may affect counts differently from averages. Choose the intended function and verify its valid observations. Do not assume that a successfully created table contains the requested statistic.

Worked example: eight clinic encounters

The supplied teaching example contains three Urgent waits of **12, 18 and 15 minutes**, three Priority waits of **28, 34 and 31 minutes**, and two Routine waits of **45 and 52 minutes**. It represents a small morning intake sample. These values illustrate grouping; they are not the Metro delivery-lab dataset.

For Urgent, the sum is 45 minutes across three encounters, giving a mean of $45 \div 3 = 15$ **minutes**. Deviations from 15 are -3, 3 and 0; their squares sum to 18. Sample variance is $18 \div (3 - 1) = 9 \text{ min}^2$. Sample **standard deviation (SD)**, the square root of variance, is **3 minutes**.

For Routine, the mean is $(45 + 52) \div 2 = 48.5$ **minutes**. Both observations are 3.5 minutes from that mean. Squared distances sum to 24.5 min^2 , which is divided by one because the sample count is two. The resulting sample SD is **4.95 minutes**, rounded to two decimal places.

Group	Valid count, n	Mean wait (minutes)	Sample SD (minutes)
Urgent	3	15.00	3.00
Priority	3	31.00	3.00
Routine	2	48.50	4.95
All encounters	8	29.38	14.24

Build the example PivotTable by putting Triage_Level in Rows and Wait_Minutes in Values three times. Set the three summaries to Count, Average and **StdDev**, Excel's sample-standard-deviation label. Standard PivotTable summary functions do not supply ordinary median and quartile calculations. Use the supplied formulas when those are required, as they are in the Metro lab.

Routine has the highest mean in this sample. That describes the recorded encounters; it does not establish poor performance. Groups differ in urgency, counts are tiny, and arrival conditions or staffing may differ. A higher wait could reflect the prioritization process rather than a failure. Keep that interpretation separate from any decision about service quality.

GO DEEPER Deeper explanation: why the grand mean needs weights

A **weighted mean** gives each group mean influence in proportion to its valid count. Let n_g be the valid count in group g and $\bar{x}_g$ its mean. The overall mean is the sum of each count multiplied by its group mean, divided by the total valid count. Here:

$$\begin{aligned}\text{Total wait} &= (3 \times 15) + (3 \times 31) + (2 \times 48.5) = 235 \text{ minutes} \\ \text{Overall mean} &= 235 / (3 + 3 + 2) = 29.375 \approx 29.38 \text{ minutes}\end{aligned}$$

The symbol $\approx$ means approximately equal after rounding. Simply averaging the three group means gives 31.5 minutes. That incorrectly gives the two Routine encounters the same combined weight as either three-encounter group. Weighting restores the arithmetic contribution of each observation.

The overall SD is not an average of group SDs either. It includes both variation inside each group and separation between group centers. Here the within-group SDs are small, yet the overall SD is much larger because the group means differ. Use the underlying observations or a valid pooled-variance calculation, which combines both components. Keep that extended formula in the full toolkit when needed.

Compare groups without assigning every person the average

Within-group variation describes differences among members of one group. **Between-group differences** compare group summaries. Both matter. A group's mean does not describe every member, and overlapping distributions can make an apparent ranking less informative for individual cases.

An **ecological fallacy** assigns a group-level relationship to individual members without supporting evidence. A branch's higher average sale does not mean every transaction there exceeds every transaction elsewhere. A **composition effect** occurs when differences in the types or proportions of cases change an aggregate. A clinic serving more urgent patients may have a different overall wait even if each urgency group's process is unchanged.

A **confounder** is another variable related to both the group being compared and the outcome, which can distort a causal interpretation. **Stratification** means examining meaningful subgroups separately. It can clarify composition, but splitting into very small groups leaves weak evidence. Choose subgroups for a reason grounded in the question or workflow, not because they produce a preferred conclusion.

Simpson's paradox describes cases where a relationship within groups weakens or reverses after groups are combined. It is a reminder to compare overall and subgroup results with their counts. It is not evidence that every aggregate is misleading. Weighting fixes arithmetic; it does not itself resolve confounding or unequal data quality.

Follow the grouped-analysis workflow

1. Declare the grain and eligible observations. Verify identifiers at that level.
2. Check group labels for inconsistent spelling, spaces and missing categories.
3. Apply the measurement's own validity rule and confirm units.
4. Calculate valid count, meaningful center and spread for every group.
5. Check that subgroup counts reconcile with the overall count when groups are exhaustive and do not overlap.
6. Compare distributions on a shared scale; investigate unusual observations without automatically deleting them.
7. Write one finding, its decision relevance and a limitation. Save the calculation steps as well as the output.

An analysis is **reproducible** when another analyst can repeat it using the same source and documented decisions. A screenshot records an output, but not every filter, formula or setting that produced it. Keep the workbook formulas and supplied syntax with the stated file version and missing-value rules.

Bring the sections together in Part D

Open **IN-CLASS_Lab Part D**. In **B7:F10**, calculate valid counts, medians, first quartiles, third quartiles and interquartile ranges for the hubs and all-hubs row. The **interquartile range (IQR)** is the third quartile minus the first, describing middle-half width. These are formula-based summaries; do not substitute a PivotTable average for a median.

The input cells and adjacent checks guide the calculation. Confirm that hub counts sum to the all-hubs valid count and that each IQR equals its two quartiles' difference. Complete the all-hubs row separately. Its median and IQR are calculated from the combined valid observations, not from an average of hub medians or IQRs.

Continue the existing Part D written work and **H4** SPSS evidence during the guide's take-home checkpoint, then follow its next instructions. Parts A, B and C return during the scheduled evidence clinic. The Mac screenshots and exact clicks remain in **LAB01_Student_Lab.pdf** or its HTML edition. This concise toolkit is study support, not an additional graded worksheet.

Reflect: Penalizing a carrier now may improve accountability but assign blame incorrectly. Waiting for comparable records improves the evidence while Metro pays interim support costs. Which consequence is acceptable, and what new evidence would change your choice?

Recall without looking back: Why is the simple average of unequal-sized group means usually wrong? Why does a correct weighted mean still need a cautious interpretation?

Lab connection: Save your personalized workbook, complete the existing chart-supported claim and explanation, and follow the guide's submission checklist. A finished table is an output; an evidence-supported decision is the purpose.